

# Nathan Genetzky's Resume

Software and Hardware

Nathan Genetzky

2026-09-22

## Career Profile

- Embedded Linux engineer with ~9 years taking connected products from board bring-up through secure, field-updatable deployment.
- Yocto distributions, BSP and kernel work, CI/CD, and hardware-in-the-loop test.
- Product cybersecurity (EN 18031 / EU RED, secure boot, SBOM) and OTA update systems.
- Electrical Engineering graduate with honors distinction.

## Contact

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## Experiences

**Senior Software Engineer** PTx, AGCO's Precision Agriculture Technology business (04.2024 - PRESENT)

- Develop embedded Linux platforms for agricultural equipment electronics; working knowledge of CAN bus and protocol.
- Applied EN 18031 (EU RED cybersecurity) to a connected product: asset identification, decision-tree assessment, and evidence for third-party certification; tracking CRA readiness.
- Implemented secure boot and signed firmware images; performed threat modeling and key management backed by a hardware security element.
- Generate and maintain SBOMs for a Yocto-based product.

- Implemented RAUC OTA updates across multiple A/B and asymmetric partition layouts: U-Boot integration, secure boot fallback handling, recovery from failed kernel, rootfs, and application updates, and the tests to prove it.
- Build and deliver containerized applications that run on embedded targets.
- Integrated AI-assisted development into daily embedded workflow for code generation, review, test authoring, and documentation, with defined boundaries on where generated code is not trusted without verification.

**Senior Software Engineer** ESW Team at Appareo Systems (acquired by AGCO, 01.2022) (12.2021 - 04.2024)

- Performed board bring-up for custom PCB with heterogeneous SOC (IMX8DXL), debugging with JTAG/SWD, GDB, logic analyzer, and oscilloscope.
- Integrated drivers for cellular modems, WiFi, and bluetooth.
- Setup dockerized Jenkins with manual and GitOps centric builds, using Gitflow with automated build and deployment of releases.
- Developed and utilized HIL testing with Labgrid on RPI4.
- Generated SDKs and docker images for quick cross-compilation of C++ Applications.

**Software Engineer 1** System Team at Vaddio (05.2017 - 12.2021)

- Working with FPGA and python developers to create an integrated embedded system that routes audio and video signals in professionally built AV systems.
- I help to build a custom Linux distribution from the ‘/’ up with the Yocto Project.
- Our team uses many other tools during development such as Jenkins, Jira, Git, Bitbake, and Puppet.

**Software Engineer** Dojo Five (11.2018 - 12.2021)

- Develop embedded software for Particle, Nordic, and STM32 microcontrollers
- Utilized custom hardware with the Particle platform for a custom door access solution
- Developed with NRF52 SDK to create devices in a complex robotic system with BLE services

**Embedded Linux and OTA Experiments** Personal projects, self-directed (ONGOING)

- Compared Mender and OSTree against RAUC to understand A/B, atomic, and rollback-safe update strategies, and authored a custom update agent to work through the trade-offs first hand.

- Managed device fleets with Balena: provisioning, remote access, and delivery of containerized applications to embedded targets.

## Projects

### Hardware Integrated Prototyping Environment at SDSU Engineering Expo

- Nathan Genetzky, Jordan Ulmer, Tanner Johnson (SDSU, 2017)
- <http://academic.genetzky.us/project/sdsu/2017-hipe/>

### Design and Verification of a SPI to JTAG Interface Adapter

- Nathan Genetzky, Jordan Ulmer (SDSU, 2017)
- <http://academic.genetzky.us/publication/sdsu/2017-05-05-ee492-spi2jtag/>

### Interactive User Interface with PIC18 Microcontroller

- Nathan Genetzky, Drake Jenö (SDSU, 2017)
- <http://academic.genetzky.us/publication/sdsu/2015-05-01-ee347-interactive-ui-with-pic-microcontroller/>

### Particle Projects

- Firmware for micro controllers sold by particle.io for use with wifi or cellular cloud devices.
- <http://academic.genetzky.us/project/particle-projects/>

### Skills & Proficiency

- Yocto (Build System, Embedded Distro, BSP Support) (90%)
- Developer Tools (Docker, Jenkins, Git, GDB, JTAG/SWD) (85%)
- Terminal Workflow (Vim, GNU tools, shell scripting) (85%)
- Linux Kernel (Drivers, Config, BSP) (55%)
- Python Middleware (DBus, REST, GObject, threading) (35%)
- FPGA Development (SW Interface, Reusable Verilog, Automated Builds) (25%)

### Education

**BS Electrical Engineering** South Dakota State University (2012 - 2017)

**Minor Software Engineering** South Dakota State University (2012 - 2017)

## **Lauguages**

- Bash (~10 years)
- Python (~8 years)
- C++ (~9 years)
- C (~7 years)
- Dockerfile (~5 years)
- Java for Android (~1 year)

## **Interests**

- hobby-electronics, home automation, embedded-linux
- ice and rock climbing
- snowboarding and skiing
- gardening, cooking
- jetsking, boating, water skiing
- camping, kayaking, volleyball